



The quant platform that tells you the truth.

Project Overview & Feature Summary

An honest quantitative trading research platform that helps everyday traders make disciplined, evidence-based decisions - and tells them the truth about whether a strategy actually works.

Shared for review - runs out of the box in demo mode (synthetic data, no API keys).

1. What AlphaForge is

AlphaForge takes the tools professional quant desks use - signal generation, rigorous backtesting, risk analytics, portfolio construction - and packages them for ordinary traders, with two things almost no retail tool offers: an engine that refuses to lie about a strategy's real edge, and an AI layer that lets a non-expert drive the whole thing in plain English.

It is a research and decision tool, not a magic profit machine. The promise is honesty and discipline: test an idea properly, then get a blunt verdict on whether the edge is probably real or probably an overfit illusion - before any real money is involved.

2. The problem it solves

Most ordinary traders lose money for boring, fixable reasons - not for lack of a crystal ball:

- They trade on emotion, tips and hunches instead of tested rules.
- They never honestly test a strategy, or fool themselves with a backtest that secretly peeks at the future.
- They ignore transaction costs (brokerage, STT, GST, slippage) that quietly eat returns.
- They have no risk limits, so a single bad streak wipes out the account.

This is a discipline and honesty problem - and software is genuinely good at enforcing discipline and honesty. That is the gap AlphaForge fills.

3. What makes it different

THE UNIQUE SELLING POINT

"The platform that tells you the truth."

Most retail apps either draw a pretty equity curve going up (and quietly lie), or sell black-box signals with no transparency. AlphaForge is built on the opposite premise: every result is stress-tested for luck and overfitting, and the numbers you see are numbers you can trust.

4. Key features

THE MOAT

Honesty Engine - overfitting control

Every backtest is graded with the academic overfitting toolkit - the Deflated Sharpe Ratio and the Probability of Backtest Overfitting - and returns a blunt verdict: 'probably real' vs 'likely overfit - do not trade it'. It also answers what users actually care about: would this have blown up my account, and does it beat simply buying and holding after costs?

ACCESSIBILITY

AI Layer - plain English to tested strategy

Describe a strategy in ordinary words ('buy when RSI drops below 30, sell at 70') and AlphaForge turns it into a runnable, tested rule. It then explains the result conversationally, acts as an AI risk auditor that warns when a strategy looks overfit or dangerous, and grounds its advice in real quant-finance research (with sources cited) rather than hallucinating. Works offline; upgrades to richer explanations when an AI key is added.

FORWARD PROOF

Paper Trading - forward validation

Deploy a strategy to a paper account and track it forward, bar by bar, on data it has never seen. This closes the loop between 'the backtest looked great' and 'it actually works' - with simulated money only, before any real capital is ever considered.

FOUNDATION

Correct-by-construction analytics

A vectorised backtest engine that avoids lookahead bias by design, models real India and US transaction costs, and validates with walk-forward analysis and Monte Carlo simulation. Backed by a full metrics library (Sharpe, Sortino, drawdown, VaR incl. Student-t and GARCH), technical indicators, a multi-signal alpha engine, factor analytics, HMM regime detection and portfolio optimisation.

ACCESS

Web app + REST API

A multi-page web application (Dashboard, AI Strategy Lab, Backtest, Paper Trading, Risk, Portfolio, Factors, Regime, Signals, Alerts, Prediction and more) plus a clean REST API of ~32 endpoints wrapping the core, so the whole platform can be driven programmatically.

5. Current status

A substantial, working codebase (~24,000 lines of Python) with the hard, correctness-critical parts in place and a growing automated test suite (200+ tests passing).

Area	Status	Notes
Data, metrics, indicators	Done	Clean OHLCV engine, fat-tail & GARCH risk, demo-mode synthetic data
Backtest engine	Done	No lookahead, real cost models, walk-forward, Monte Carlo
Factor / regime / portfolio	Done	IC analytics, HMM regimes, optimisation
Honesty Engine (DSR / PBO)	Done	The key differentiator - verdict on every backtest
AI Layer	Done	NL to strategy, explainer, grounded risk auditor
Paper trading	Done	Local simulator + optional broker paper account
REST API + Web UI	Done	~32 endpoints; multi-page app
3D web frontend	Planned	Immersive Next.js / Three.js experience (in design)

The "3D web frontend" row is a planned redesign of the user interface only - an immersive, animated, physics-driven experience. It does not change any of the quant logic above.

6. Technology

- **Core / backend:** Python, NumPy, pandas, SciPy, statsmodels, arch (GARCH), scikit-learn.
- **App & API:** Streamlit multi-page app, FastAPI REST service, Plotly charts.
- **AI layer:** plain-English parser + retrieval over a curated quant-finance knowledge base; optional LLM for conversational output.
- **Planned frontend:** Next.js, React, Three.js (react-three-fiber), physics & GLSL shaders.

7. What we'd love your feedback on

- Is the core promise - **honesty over hype** - clear and compelling?
- Does 'describe a strategy in plain English and get an honest verdict' sound genuinely useful to you?
- Which feature stands out most - the Honesty Engine, the AI layer, or paper-trading validation?
- Would an immersive 3D interface make you more likely to try and trust the product?

Disclaimer: AlphaForge is an educational research tool, not financial advice. It grades strategies for honesty - it does not predict the future. Demo mode uses synthetic data and no real money is ever involved.